UE24CS352A | MACHINE LEARNING | MINI-PROJECT REPORT

A Proximity-Based Early Warning System for Gentrification in California

Binary classification of California Census Tracts using housing, socioeconomic and spatial proximity features

COURSE

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Experimental proof of concept. Not intended for operational or policy use.

Abstract

Gentrification is a gradual process in which housing costs and the socioeconomic profile of a neighbourhood change. This project asks whether historical housing and socioeconomic indicators, combined with four spatial proximity features, can help classify California Census Tracts as gentrifying or not gentrifying. The prepared dataset contains 7,965 tracts and 154 model input features, and the binary target is `housing_gentrification_label`. Logistic Regression, Random Forest, a baseline XGBoost model and a tuned XGBoost model were compared on predefined training, validation and test partitions.

On the validation partition (398 tracts) the Random Forest performed best, with accuracy 0.5754, F1 score 0.6269 and ROC-AUC 0.5937. Removing the four proximity features lowered its validation ROC-AUC to 0.4888. The selected Random Forest was then retrained on the combined training and validation data (7,566 tracts) and evaluated on the 399-tract test partition, where it reached accuracy 0.4912, F1 score 0.5519 and ROC-AUC 0.5125, which is close to random discrimination. The validation advantage therefore did not carry over to unseen data.

We present the work as an experimental proof of concept rather than a usable early-warning system. The report also discusses its main limitations, including that the procedure used to construct the target label is not documented in the project files, and it suggests realistic next steps.

1. Introduction and Background

Gentrification generally refers to neighbourhood change in which rising housing costs and a shifting resident profile alter the character of an area. It is usually discussed together with concerns about housing affordability and the displacement of existing residents. Being able to recognise early which areas are likely to experience such change would be of interest to planners, housing agencies and community organisations, because it could allow attention and resources to be directed before change is well advanced.

Predicting this kind of change is difficult. Neighbourhood change unfolds slowly, is measured only through coarse survey and administrative data, and depends on many factors that are not captured in any single table. In this project the unit of analysis is the Census Tract, a small statistical area defined by the U.S. Census Bureau, and each tract is described by a large set of tabulated indicators.

Neighbouring tracts are not independent of one another. Nearby places tend to resemble each other, an idea often summarised as the first law of geography (Tobler, 1970). A tract surrounded by high-cost, rapidly changing tracts may therefore differ from a similar tract in a quiet area. This motivates the project's central idea of adding *proximity features*, which summarise conditions in the surrounding tracts, to the information describing each tract itself.

This report documents a machine-learning mini-project for the course UE24CS352A – Machine Learning at PES University. It describes the data, the models, the experiments and the results, and it is deliberately candid about the fact that the final test performance is weak.

2. Problem Statement and Research Question

The task is formulated as supervised binary classification. For each California Census Tract i we have a vector of input features and a binary label y in $\{0, 1\}$, where 0 denotes **Not Gentrifying** and 1

denotes **Gentrifying**. The label used is the column `housing_gentrification_label` of the prepared dataset. The goal is to learn a function that predicts this label from the input features.

The research question is:

> **Can historical socioeconomic and housing features, together with proximity-based spatial information, provide useful predictive signals for identifying Census Tracts associated with gentrification in California?**

Three supporting questions follow from it. First, which of the candidate models discriminates best on held-out validation data? Second, do the four proximity features contribute predictive information? Third, does the performance observed on validation data persist on a separate test partition?

3. Project Objectives and Scope

3.1 Objectives

1. Prepare and analyse a California Census Tract dataset for binary gentrification classification.
2. Compare Logistic Regression, Random Forest, XGBoost and a tuned XGBoost model.
3. Investigate, through an ablation experiment, whether the proximity features add predictive information.
4. Evaluate the models with accuracy, precision, recall, F1 score and ROC-AUC.
5. Select a model on validation data and measure its generalisation on a separate test partition.
6. Analyse the limitations of the approach and identify directions for improvement.

3.2 Scope

The project is limited to training and evaluating classifiers on a prepared, tract-level dataset using its predefined partitions. It does **not** reconstruct how the target label was generated, make causal claims about gentrification, forecast change for future years, use spatially or temporally blocked validation, or build a deployable system. These boundaries matter when reading the results in later sections.

4. Dataset

4.1 Overview

The data are provided as a single prepared file, `california_gentrification_final_model_ready.csv`, with one row per Census Tract. The original source of the underlying data and the steps used to prepare the file are not documented in the project notebook, so this report describes the dataset only through its structure and column names. Table 1 summarises its main properties.

Table 1. Summary of the prepared dataset.

Property	Value
Observations (Census Tracts)	7,965
Total columns	167 (160 float, 6 integer, 1 text column)
Distinct county codes (`county_fips`)	58
Machine-learning input features after exclusions	154
Of which: tract-level change indicators / proximity features	150 / 4
Missing values in the 154 input features	0
Missing values in the full file (excluded columns only)	553 (546 in `pre_housing_change`, 7 in `pre_income_change`)
Duplicate rows	0

4.2 Target distribution and data partitions

The target `housing_gentrification_label` is binary with no missing values. Class 0 (Not Gentrifying) contains 3,833 tracts (48.12%) and class 1 (Gentrifying) contains 4,132 tracts (51.88%), so the classes are close to balanced. Because of this balance, accuracy is a meaningful metric here, although a trivial classifier that always predicts the majority class still reaches roughly 52% on every partition (Section 11).

The dataset carries a predefined `split` column that assigns each tract to the training, validation or test partition. The way this split was generated is not documented in the notebook. A notebook check found that 99% of test tracts belong to a county that also appears in the training partition, so the split does not hold out whole regions. Table 2 gives the partition sizes and class shares.

Table 2. Partition sizes and class balance.

Partition	Tracts	Class 1 (Gentrifying)	Class 1 share
Training	7,168	3,729	52.02%
Validation	398	210	52.76%
Test	399	193	48.37%
Total	**7,965**	**4,132**	**51.88%**

The validation and test partitions are small, with fewer than 400 tracts each. This limits how precisely any metric can be estimated, a point that becomes important when the results are interpreted. The test partition also has a somewhat lower share of gentrifying tracts than the other two.

4.3 Feature preparation

The feature matrix was built by removing 13 of the 167 columns from the file, leaving 154 inputs. The excluded columns and the reasons recorded in the notebook are listed in Table 3. The six target-related columns were removed explicitly to avoid leakage of outcome information into the model inputs, and a notebook audit confirmed that none of them, nor the target itself, remains among the inputs.

Table 3. Columns excluded from the model inputs.

Group	Columns	Reason
Identifiers (4)	`tract_id`, `state_fips`, `county_fips`, `tract_code`	Identify a tract or area rather than describe it.
Target and target-related (6)	`housing_gentrification_label` (target), `housing_gentrification_score`, `housing_cost_response`, `income_gentrification_label`, `income_gentrification_score`, `income_hellinger_response`	Outcome information; using it as input would leak the answer.
Incomplete change variables (2)	`pre_housing_change`, `pre_income_change`	Contain missing values (546 and 7).
Partition indicator (1)	`split`	Used only to assign rows to partitions.

The 154 remaining features consist of 150 columns whose names begin with `delta_2010_2011_` followed by a table and variable code (for example `delta_2010_2011_DP4_HC01_VC71`), plus the four proximity features described in Section 5. The prefix suggests that the 150 columns are differences between 2010 and 2011 values of tract-level housing, social, economic and demographic indicators, and the codes resemble the labelling of American Community Survey profile tables. The notebook does not document this, so we treat the interpretation as an inference from the column names only. No further feature engineering, imputation or selection was applied. The Logistic Regression model standardises the features inside its pipeline, while the tree-based models use the raw values.

5. Proximity-Based Features

5.1 Purpose

The four proximity features give each tract information about its surroundings. They allow a model to take into account not only a tract's own characteristics but also whether neighbouring tracts have high or low values of the same quantity, and whether the tract stands out from them. The features are named `spatial_lag_housing_pre2012`, `local_morans_I_housing_pre2012`, `spatial_lag_income_pre2012` and `local_morans_I_income_pre2012`.

5.2 General definitions

This subsection gives the standard definitions of the two statistics. Section 5.3 explains what can and cannot be confirmed about the columns used in this project.

Spatial lag. The spatial lag of a variable x at location i is a weighted average of the values of x at the neighbouring locations:

$$(Wx)_i = \Sigma_j \, w_{ij} \, x_j$$

Here w is a weight that is positive when tract j counts as a neighbour of tract i and zero otherwise. With row-standardised weights, which sum to one for each tract, the spatial lag is simply the mean value of the neighbours. A high spatial lag means that the surroundings have high values of the variable.

Local Moran's I. Local Moran's I is a local indicator of spatial association (Anselin, 1995). For location *i* it can be written as

$$I_i = (z_i / m_2) \Sigma_j w_{ij} z_j$$

where z is the deviation of the variable from its overall mean and m_2 is the average of the squared deviations. A large positive value indicates that a location and its neighbours deviate from the mean in the same direction, so that similar values are clustered (high values among high values, or low among low). A negative value indicates a location that differs from its neighbours. Because it combines a tract's own deviation with that of its neighbours, it captures local clustering rather than only the neighbourhood average.

5.3 What is and is not verified for this project

The four proximity columns are present in the prepared dataset as ready-made values; the notebook does not compute them. The following can be confirmed from the notebook and the file: the four columns exist, they are included among the 154 input features, and they were removed in the ablation experiment. The column names indicate that two of the features concern housing and two concern income, that each variable has a spatial lag and a Local Moran's I version, and that the underlying values refer to the period before 2012.

The following details are **not** documented in the project files and are therefore not claimed in this report: how neighbours were defined (for example by shared borders, distance or nearest neighbours), the weighting scheme, the exact variables used, the precise time period, and which tracts were used when computing the statistics. The general definitions above describe what these statistics measure; they should not be read as a description of how the supplied columns were produced.

6. Methodology and Workflow

The workflow was implemented in Python in a Google Colab notebook, using pandas and NumPy for data handling, scikit-learn for preprocessing, models and metrics, XGBoost for gradient boosting, and Matplotlib for the figures. The steps were as follows.

1. **Load and audit the data.** The CSV file was loaded and checked for dimensions, data types, missing values, duplicates and the target distribution.
2. **Build the feature matrix.** The 13 excluded columns (Table 3) were removed, leaving 154 features, and the target was separated. A leakage audit confirmed that the target and the target-related columns were not among the inputs.
3. **Apply the predefined partitions.** The `split` column was used to form the training (7,168), validation (398) and test (399) sets.
4. **Train candidate models on the training partition.** Logistic Regression, Random Forest, baseline XGBoost and, after a randomised search, tuned XGBoost were fitted (Section 7).
5. **Compare candidates on the validation partition.** Accuracy, precision, recall, F1 score and ROC-AUC were computed for every model.
6. **Run the proximity ablation.** The Random Forest and the baseline XGBoost were retrained without the four proximity features (150 inputs) and evaluated on the validation partition.
7. **Examine feature importance.** Built-in importances and permutation importance were inspected (Section 10).

8. **Select and retrain the final model.** Random Forest was selected on the basis of validation ROC-AUC and F1 score, retrained on the combined training and validation data (7,566 tracts) and evaluated on the test partition (399 tracts).

A fixed random seed of 42 was used wherever a model has a random component. Each configuration was run once, so the results do not include repeated runs or confidence intervals. In this report, Sections 8 to 11 present measured results, and Section 12 gives our interpretation of them.

7. Model Configurations and Evaluation Metrics

7.1 Models

Logistic Regression is a linear classifier that estimates the probability of the positive class and serves as a simple baseline. Random Forest (Breiman, 2001) averages many decision trees trained on random subsets of the data and features, and can capture non-linear relationships and interactions. XGBoost (Chen and Guestrin, 2016) builds trees sequentially, each one correcting the errors of the previous ones. Table 4 lists the recorded configurations. Parameters that are not listed were left at the library defaults.

Table 4. Model configurations used in the experiments.

Model	Configuration
Logistic Regression	`StandardScaler` followed by `LogisticRegression(max_iter=2000, random_state=42)`, as a scikit-learn pipeline
Random Forest	`n_estimators=300`, `random_state=42`, `n_jobs=-1`
XGBoost (baseline)	`n_estimators=300`, `max_depth=5`, `learning_rate=0.05`, `subsample=0.8`, `colsample_bytree=0.8`, `random_state=42`, `eval_metric='logloss'`
XGBoost (tuned)	Randomised search (12 sampled configurations, 3-fold cross-validation, ROC-AUC scoring, `random_state=42`) fitted on the training partition only. Best parameters: `subsample=0.9`, `reg_lambda=5`, `n_estimators=300`, `min_child_weight=5`, `max_depth=4`, `learning_rate=0.08`, `colsample_bytree=1.0`. Best cross-validation ROC-AUC: 0.5389
Ablation variants	Random Forest and baseline XGBoost with the same settings, trained on the 150 features that remain after removing the four proximity features

The search space for the tuned XGBoost model is shown in Table 5. Randomised search samples a limited number of configurations from this space instead of trying every combination (Bergstra and Bengio, 2012). Only XGBoost was tuned; the Random Forest and Logistic Regression use the configurations in Table 4.

Table 5. Search space for the randomised XGBoost tuning.

Parameter	Values searched
`n_estimators`	200, 300, 400, 500
`max_depth`	3, 4, 5, 6
`learning_rate`	0.03, 0.05, 0.08, 0.1
`subsample`, `colsample_bytree`	0.7, 0.8, 0.9, 1.0 (each)
`min_child_weight`	1, 3, 5
`reg_lambda`	1, 5, 10

7.2 Evaluation metrics

All metrics treat **Gentrifying** (class 1) as the positive class. A *true positive* (TP) is a gentrifying tract predicted as gentrifying, a *false positive* (FP) is a non-gentrifying tract predicted as gentrifying, a *true negative* (TN) is a non-gentrifying tract predicted as non-gentrifying, and a *false negative* (FN) is a gentrifying tract predicted as non-gentrifying. Table 6 defines the five metrics.

Table 6. Evaluation metrics.

Metric	Definition	Plain-language reading
Accuracy	(TP + TN) / all predictions	Share of tracts classified correctly.
Precision	TP / (TP + FP)	Of the tracts flagged as gentrifying, the share that really are.
Recall	TP / (TP + FN)	Of the truly gentrifying tracts, the share that were flagged.
F1 score	2 × precision × recall / (precision + recall)	Harmonic mean of precision and recall; low if either is low.
ROC-AUC	Area under the ROC curve	Chance that a randomly chosen gentrifying tract receives a higher score than a randomly chosen non-gentrifying tract.

Accuracy, precision, recall and F1 score depend on the decision threshold used to turn predicted probabilities into class labels (the default threshold of 0.5 was used throughout). ROC-AUC does not depend on a single threshold because it evaluates how well the scores rank the two classes (Fawcett, 2006). A ROC-AUC of 0.50 corresponds to random ranking, a value of 1.0 to perfect ranking, and values only a little above 0.50 indicate discrimination that is close to random. ROC-AUC was given particular weight in model comparison for this reason.

8. Validation Results

Table 7 reports the validation results. Every model was trained on the training partition and evaluated on the 398 validation tracts. The two bottom rows are the ablation variants without the four proximity features.

Table 7. Validation results (398 tracts; models trained on the training partition).

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Random Forest	**0.5754**	**0.5844**	**0.6762**	**0.6269**	**0.5937**
Logistic Regression	0.5101	0.5301	0.6286	0.5752	0.5226
XGBoost	0.5302	0.5535	0.5667	0.5600	0.5302
Tuned XGBoost	0.5276	0.5514	0.5619	0.5566	0.5195
XGBoost without proximity features	0.4874	0.5143	0.5143	0.5143	0.4717
Random Forest without proximity features	0.4950	0.5181	0.6143	0.5621	0.4888

Random Forest obtained the highest value on every metric in the table and was selected for the final evaluation, mainly because it had the strongest validation ROC-AUC (0.5937) and F1 score (0.6269). Baseline XGBoost (0.5302), Logistic Regression (0.5226) and tuned XGBoost (0.5195) all had validation ROC-AUC values close to 0.50.

Tuning did not improve XGBoost in this experiment. The best cross-validation ROC-AUC found during the search was 0.5389, but the tuned model scored 0.5195 on the validation partition, which is lower than the 0.5302 of the baseline configuration. Figure 1 compares the validation ROC-AUC values of the five main configurations.

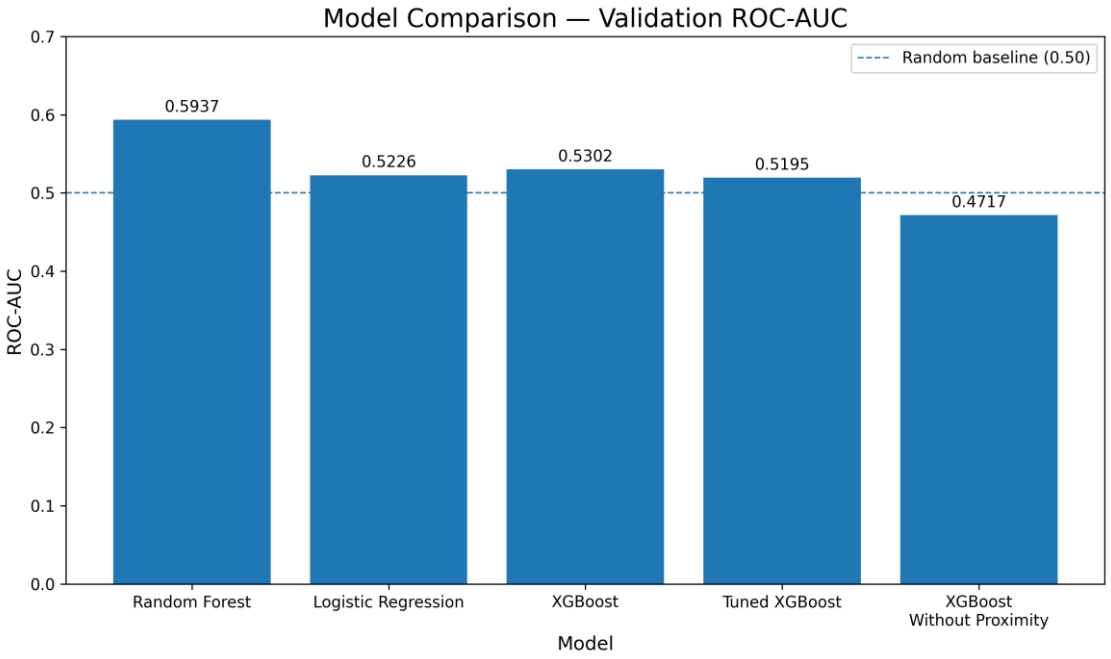


Figure 1. Validation ROC-AUC of the candidate models. The dashed line marks ROC-AUC = 0.50, which corresponds to random discrimination. The Random Forest without proximity features (0.4888) is not part of this chart; it is shown in Figure 2 and listed in Table 7.

9. Proximity-Feature Ablation

To examine the role of the proximity features, the Random Forest was retrained with the four features removed and compared with the full model on the validation partition. The same comparison was made for baseline XGBoost. The results are in Table 8 and, for the Random Forest, in Figure 2.

Table 8. Effect of removing the four proximity features (validation ROC-AUC).

Model	With proximity (154 features)	Without proximity (150 features)	Difference
Random Forest	**0.5937**	**0.4888**	**0.1049**
XGBoost (baseline)	0.5302	0.4717	0.0585

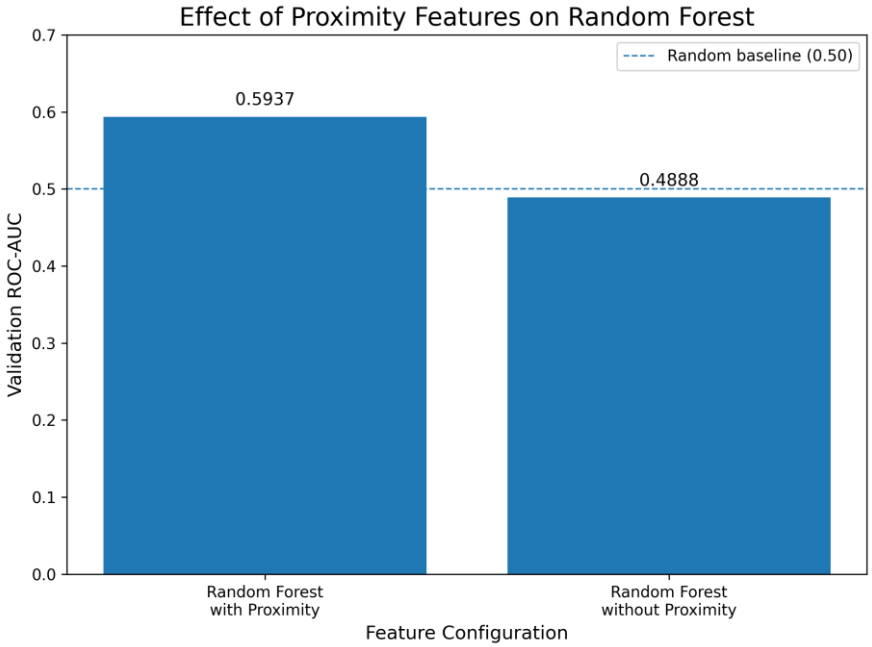


Figure 2. Validation ROC-AUC of the Random Forest with all 154 features (including the four proximity features) and with the 150 features that remain after removing them. The dashed line marks ROC-AUC = 0.50 (random discrimination).

Removing the proximity features reduced the Random Forest validation ROC-AUC by 0.1049, from 0.5937 to 0.4888, and also lowered its accuracy (0.5754 to 0.4950) and F1 score (0.6269 to 0.5621). The baseline XGBoost model showed the same direction of change, with a smaller drop of 0.0585. In this experiment, therefore, models that could use the proximity features ranked the validation tracts better than models that could not.

This is an observed predictive association within one experiment. It does not show that proximity causes gentrification, and it does not show that the features will improve performance on other data. The comparison rests on a single run with a fixed random seed, on a validation set of 398 tracts, and was not repeated on the test partition. These caveats are taken up in Section 12.

10. Feature Importance

Feature importance was examined with three tools in the notebook. These indicate which inputs a model relied on; they do not indicate that a feature causes gentrification.

10.1 Random Forest built-in importance

The built-in (impurity-based) importance of the Random Forest trained on the training partition ranked the four proximity features as the four most important of the 154 inputs: `spatial_lag_housing_pre2012` (0.0165), `local_morans_I_housing_pre2012` (0.0139), `spatial_lag_income_pre2012` (0.0124) and `local_morans_I_income_pre2012` (0.0102). The fifth-ranked feature scored 0.0087. All importance values are small, because the importance is spread over many features.

10.2 XGBoost built-in importance

For the baseline XGBoost model, `spatial_lag_housing_pre2012` and `local_morans_I_housing_pre2012` ranked first and second among the top 20 features, and all four proximity features appeared in that top 20 (Table 9).

Table 9. Top-ranked features by built-in XGBoost importance (baseline XGBoost).

Rank	Feature	Importance
1	spatial_lag_housing_pre2012	0.0114
2	local_morans_I_housing_pre2012	0.0090
3	delta_2010_2011_DP2_HC01_VC182	0.0087
4	delta_2010_2011_DP4_HC01_VC71	0.0085
5	delta_2010_2011_DP5_HC01_VC47	0.0081
6	spatial_lag_income_pre2012	0.0080
11	local_morans_I_income_pre2012	0.0075

10.3 Permutation importance

Permutation importance measures how much a model's score falls when the values of one feature are randomly shuffled. It was computed for the baseline XGBoost model on the validation partition, using ROC-AUC as the score with 10 repetitions. The highest-ranked feature was `local_morans_I_housing_pre2012`, with a mean decrease in ROC-AUC of approximately 0.0237, followed by `spatial_lag_income_pre2012` at approximately 0.0115 (Table 10). The other two proximity features did not appear in the top 20 of this ranking.

Table 10. Top-ranked features by permutation importance (baseline XGBoost, validation set).

Rank	Feature	Mean decrease in ROC-AUC	Std. deviation
1	local_morans_I_housing_pre2012	0.0237	0.0112
2	spatial_lag_income_pre2012	0.0115	0.0042
3	delta_2010_2011_DP4_HC01_VC92	0.0092	0.0029
4	delta_2010_2011_DP3_HC01_VC45	0.0085	0.0030
5	delta_2010_2011_DP2_HC01_VC176	0.0073	0.0025

Three points limit what these rankings can show. First, the two XGBoost analyses do not fully agree: `spatial_lag_housing_pre2012` is first by built-in importance but outside the permutation top 20. Second, the permutation importances are small and, for the top feature, their spread across repetitions is large relative to the mean. Third, they describe the baseline XGBoost model and were computed on the same validation partition used for model comparison, not on the final Random Forest evaluated on the test set. The rankings are consistent with the ablation result in suggesting that the model made use of the proximity features, but they do not add independent evidence that the features generalise.

11. Final Test Results

The Random Forest was retrained on the combined training and validation partitions (7,566 tracts, 154 features) with the configuration in Table 4 and evaluated on the 399-tract test partition. Table 11 gives the resulting metrics, and Table 12 and Figure 3 give the confusion matrix.

Table 11. Final test results of the Random Forest (399 tracts).

Metric	Value	Percentage
Accuracy	0.4912	49.12%
Precision	0.4808	48.08%
Recall	0.6477	64.77%
F1 score	0.5519	55.19%
ROC-AUC	0.5125	51.25%

Table 12. Confusion matrix on the test partition (rows: actual class, columns: predicted class).

Actual class	Predicted Not Gentrifying	Predicted Gentrifying	Total
Not Gentrifying	71 (TN)	135 (FP)	206
Gentrifying	68 (FN)	125 (TP)	193
Total	**139**	**260**	**399**

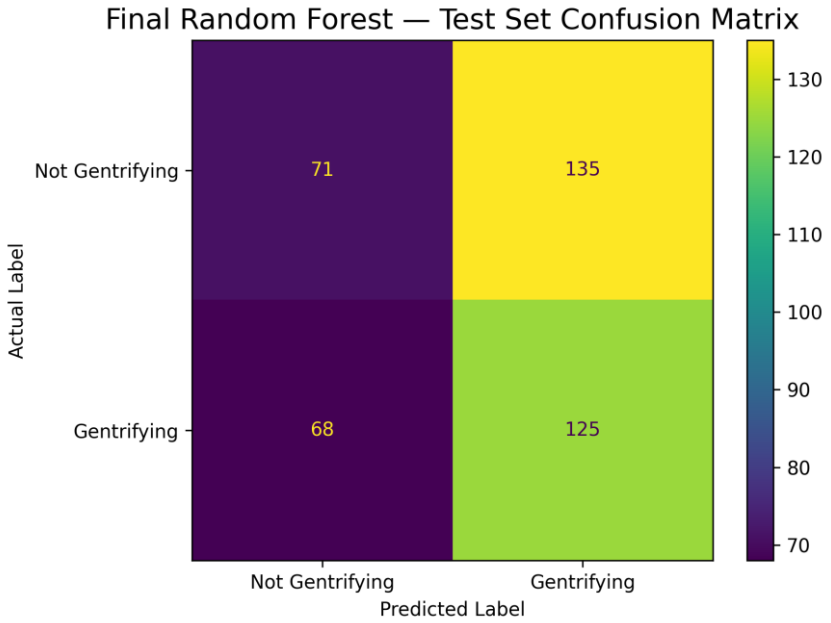


Figure 3. Confusion matrix of the final Random Forest on the 399-tract test partition. Rows show the actual class and columns the predicted class. The counts agree with Table 12.

The matrix is read as follows:

- **71** non-gentrifying tracts were correctly classified as Not Gentrifying (true negatives).
- **135** non-gentrifying tracts were wrongly predicted as Gentrifying (false positives).
- **68** gentrifying tracts were wrongly predicted as Not Gentrifying (false negatives).
- **125** gentrifying tracts were correctly classified as Gentrifying (true positives).

The reported metrics follow from these counts: accuracy is (71 + 125) / 399 = 0.4912, precision is 125 / 260 = 0.4808, and recall is 125 / 193 = 0.6477. The model also identified only 71 of 206 non-gentrifying tracts correctly, a true-negative rate of 0.3447. Table 13 sets the final test result beside the validation result of the Random Forest. The validation row refers to the model trained on the training

partition only, whereas the test model was trained on 7,566 tracts, so the two are not strictly the same model.

Table 13. Random Forest: validation and test performance.

Partition	Accuracy	Precision	Recall	F1	ROC-AUC
Validation (398 tracts)	0.5754	0.5844	0.6762	0.6269	0.5937
Test (399 tracts)	0.4912	0.4808	0.6477	0.5519	0.5125

12. Interpretation of Results

12.1 Validation versus test performance

The final test results are weak. The ROC-AUC of 0.5125 is only slightly above the 0.50 expected from random ranking, and the accuracy of 49.12% is below that of a classifier that always predicts the majority class. In the test partition 51.63% of tracts are non-gentrifying, so always predicting Not Gentrifying would have given about 51.6% accuracy. On the validation partition, by contrast, the Random Forest (0.5754) exceeded the majority-class accuracy of 52.76%. The model's precision of 0.4808 is essentially equal to the 48.37% share of gentrifying tracts in the test partition, which is the precision that would be obtained by flagging tracts at random.

Several factors probably contribute to the gap between validation and test results; the available evidence does not allow us to separate them. The validation and test partitions are small. Under the approximation of Hanley and McNeil (1982), the standard error of an AUC near 0.5 with about 200 tracts in each class is roughly 0.03 (our own calculation, not a notebook output). The test ROC-AUC of 0.5125 lies well within one standard error of 0.50, so it cannot be distinguished from chance. The validation advantage of the Random Forest over baseline XGBoost (about 0.064) is of the order of two such standard errors, and the Random Forest was chosen as the best of several candidates evaluated on the same 398 tracts, a process that tends to give the selected model an optimistic validation score. The conclusion we draw is that the model did not generalise strongly to unseen tracts, and that the validation results should not be read as a reliable estimate of future performance.

12.2 Role of the proximity features

Within the validation experiment, removing the proximity features lowered ROC-AUC by 0.1049 for the Random Forest and 0.0585 for XGBoost, and the proximity features ranked highly in the importance analyses. These observations indicate that the models used the proximity features and that, on validation data, those features carried predictive information that the other 150 inputs did not. They are not evidence of a causal effect. Note also that the Random Forest without proximity features scored below 0.50 on validation, which suggests that the remaining 150 features offered little usable signal in this setup. Because the full model also performed close to chance on the test partition, the ablation does not establish that proximity information would improve predictions in practice.

12.3 False positives and false negatives in an early-warning setting

In an early-warning application the two kinds of error carry different costs. A **false positive** flags a tract that is not gentrifying. Such errors can send limited planning, outreach or funding resources to the wrong places and may attach an unwarranted label to a neighbourhood. A **false negative** misses a tract that is gentrifying. Such errors mean that residents and organisations are not alerted, and the lead time that an early-warning system is supposed to provide is lost. Which error matters more depends on how the output would be used and on what follows a flag.

The final model produced 135 false positives and 68 false negatives. It flagged 260 of the 399 test tracts (65.2%) as gentrifying, although only 48.4% actually were, and it still missed 35% of the gentrifying tracts. Its recall of 64.77% looks moderate, but it arises largely because the model predicts the positive class often. A rule that flagged a random 65% of tracts would be expected to give a recall of about 65% and a precision of about 48%, which is close to what was observed. The model therefore does not offer a favourable trade-off between the two kinds of error. Because ROC-AUC is threshold-independent and is itself close to 0.50, changing the decision threshold would shift errors between the two types without making the ranking better.

13. Limitations

The experiment establishes how several standard classifiers behaved on one prepared dataset with fixed partitions. It does not establish that the approach works for early warning. The main limitations are the following.

1. **Weak generalisation.** The test ROC-AUC (0.5125) and accuracy (49.12%) are close to chance level, and the validation results did not carry over.
2. **Target construction is not documented.** The label `housing_gentrification_label` is used as supplied, and the notebook does not reconstruct how it was generated. An exploratory check in the notebook found that the label agrees closely with the continuous column `housing_gentrification_score` (about 99.4% agreement at a cut-off near 49.2, and 93.5% at a cut-off of 50). This is an empirical observation about the supplied file and not a documented definition. We do not claim a labelling formula, threshold or time window, and we cannot confirm that the input features precede the period in which the outcome is measured, which an early-warning claim would require.
3. **The test partition was not strictly untouched.** It was kept out of model fitting and hyperparameter search, and model selection was based on validation results. However, the notebook also contains exploratory test evaluations of a Random Forest trained on the training partition only (accuracy 0.4887, ROC-AUC 0.5033) and of a retrained baseline XGBoost (accuracy 0.5188, ROC-AUC 0.5176). The XGBoost test result is comparable to the final Random Forest, so the test data give no basis for preferring one model.
4. **Small samples and single runs.** Validation and test each contain fewer than 400 tracts, each configuration was run once with one seed, and no confidence intervals or repeated cross-validation were used. Differences of a few hundredths in ROC-AUC are therefore not reliable.
5. **Spatial dependence and undocumented proximity features.** The split does not separate regions (99% of test tracts have a county in the training data), and the construction of the proximity features is not documented. Spatial dependence between neighbouring tracts that fall in different partitions may therefore make the evaluation more or less optimistic than a spatially blocked evaluation would.
6. **Association, not causation.** The ablation and the importance analyses describe what the models used for prediction. They do not show that proximity or any other feature causes gentrification.
7. **Limited tuning and features.** Only XGBoost was tuned, with 12 sampled configurations, and the default decision threshold was used. The inputs are limited to the supplied columns, which are mostly differences between two years, and may not represent the housing-market, demographic and geographic factors that drive gentrification.

14. Conclusion

This project examined whether historical housing and socioeconomic indicators, together with four spatial proximity features, can help identify California Census Tracts associated with gentrification. Among the models compared, the Random Forest performed best on the validation partition (accuracy 0.5754, F1 score 0.6269, ROC-AUC 0.5937), and removing the proximity features reduced its validation ROC-AUC to 0.4888. The proximity features also ranked highly in the importance analyses.

However, the final Random Forest, retrained on the training and validation data, reached only 0.4912 accuracy and 0.5125 ROC-AUC on the test partition, and its confusion matrix shows many false positives alongside many missed cases. The model did not generalise strongly to unseen tracts, and its test accuracy was below that of always predicting the majority class. The work should therefore be regarded as an experimental proof of concept. It suggests that neighbourhood context is worth including in such models, but it does not demonstrate a reliable early-warning system, and the model is not suitable for real-world use in its present form.

15. Future Work

1. **Document and validate the target.** Establish and record how the gentrification label is defined, including its time window, and confirm that all inputs precede the outcome period so that the early-warning framing can be tested properly.
2. **Use stronger validation.** Evaluate with spatially blocked cross-validation (for example grouping tracts by county or by spatial blocks) and with a temporal split, and report ROC-AUC with bootstrap confidence intervals over repeated runs.
3. **Make the proximity features explicit and test their sensitivity.** Recompute the spatial lag and Local Moran's I with documented neighbour definitions, compare alternatives such as shared-border and nearest-neighbour weights, and use only training tracts when computing statistics for model fitting.
4. **Strengthen the modelling.** Tune the Random Forest as well as XGBoost, reduce redundancy among the 150 change indicators by feature selection, calibrate predicted probabilities, and choose the decision threshold according to the relative cost of false positives and false negatives.
5. **Extend the feature set.** Add longer housing and income histories and additional housing-market and demographic variables, and compare against spatially aware models such as spatial regression or graph-based approaches.
6. **Consider ethical use.** Before any practical use, review how predictions could affect residents and neighbourhoods, in particular the consequences of false positives and false negatives discussed in Section 12.3.

16. References and Project Materials

16.1 References

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16.2 Project materials

The results in this report come from the following project files.

- `data_analysis.ipynb` – Jupyter/Colab notebook containing the data audit, model training, evaluation, ablation, feature-importance and plotting code.
- `california_gentrification_final_model_ready.csv` – the prepared dataset (7,965 rows, 167 columns).
- `model_comparison_roc_auc.png` – Figure 1.
- `proximity_ablation_roc_auc.png` – Figure 2.
- `final_random_forest_confusion_matrix.png` – Figure 3.

Figures 1 to 3 are the images exported by the notebook and are reproduced without modification. The software versions used in the Colab session were not recorded in the notebook.